OPERATING SYSTEMS - UNIT 1: CPU SCHEDULING

Prof. R. Sharma | Lecture notes, B.Tech CSE Semester 5

1. Why CPU scheduling?
When several processes are in the ready queue, the short-term scheduler (CPU scheduler) picks the next process to run. Goals: maximise CPU utilisation and throughput, minimise turnaround time, waiting time and response time.

Key terms
- Arrival time (AT): when the process enters the ready queue.
- Burst time (BT): CPU time the process needs.
- Completion time (CT), Turnaround time TAT = CT - AT, Waiting time WT = TAT - BT.

2. First Come First Serve (FCFS)
- Non-preemptive. Processes run in order of arrival, like a queue at a ticket counter.
- Simple to implement with a FIFO queue.
- Problem: the convoy effect - one long process makes all short processes behind it wait.

Example: P1 (BT 24), P2 (BT 3), P3 (BT 3) arriving at time 0 in that order.
Gantt chart: P1 0-24, P2 24-27, P3 27-30. Waiting times: 0, 24, 27. Average WT = 17 ms.
If the order were P2, P3, P1 the average WT drops to 3 ms - order matters in FCFS.

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3. Shortest Job First (SJF)
- Picks the process with the smallest next CPU burst.
- Provably optimal: gives the minimum average waiting time for a given set of processes.
- Non-preemptive SJF runs the chosen job to completion.
- Preemptive version = Shortest Remaining Time First (SRTF): a newly arrived process with a shorter remaining time preempts the running one.
- Difficulty: the length of the next CPU burst is not known in advance. It is predicted with exponential averaging: tau(n+1) = alpha * t(n) + (1 - alpha) * tau(n).
- Problem: starvation of long processes. Solution: ageing.

Example (non-preemptive SJF): P1 (AT 0, BT 7), P2 (AT 2, BT 4), P3 (AT 4, BT 1), P4 (AT 5, BT 4). Order: P1 0-7, P3 7-8, P2 8-12, P4 12-16. Average WT = (0 + 6 + 3 + 7) / 4 = 4 ms.

UNIT 1: CPU SCHEDULING (contd.)

4. Round Robin (RR)
- Designed for time-sharing systems. Every process gets a fixed time quantum (time slice), e.g. 4 ms.
- If a process does not finish within its time quantum it is preempted and put at the back of the ready queue.
- Performance depends heavily on the size of the time quantum:
  very large quantum -> behaves like FCFS; very small quantum -> too many context switches.
- Rule of thumb: about 80% of CPU bursts should be shorter than the time quantum.

Example: P1 (BT 24), P2 (BT 3), P3 (BT 3), time quantum = 4.
Gantt chart: P1 0-4, P2 4-7, P3 7-10, P1 10-14, 14-18, 18-22, 22-26, 26-30.
Average WT = (6 + 4 + 7) / 3 = 5.66 ms.

Comparison
FCFS - simple, convoy effect. SJF - optimal average waiting time, needs burst prediction, may starve.
Round Robin - fair, good response time, depends on the time quantum.

Important for internals: Gantt chart numericals on all three algorithms.